

CASE REPORT

FUNCTIONAL PROGRESSION AND RETURN TO SPORT
CRITERIA FOR A HIGH SCHOOL FOOTBALL PLAYER
FOLLOWING SURGERY FOR A LISFRANC INJURYDaniel S. Lorenz, DPT, PT, LAT, CSCS, USAW¹Chad Beauchamp, PT, DPT, CSCS²

ABSTRACT

Lisfranc injuries are a challenging diagnosis for the sports physical therapist because of the lack of data on how to rehabilitate them properly. To date, the available rehabilitation literature has focused on the mechanism of injury and the conservative management of this injury. Furthermore, there is a lack of consensus on the appropriate testing and return to play criteria for an athlete recovering from this perplexing injury. This case describes a high school athlete whose primary sport was football, but was injured during wrestling. He suffered a Lisfranc injury and subsequently underwent surgical fixation. The purpose of this case report is to focus on the exercise, functional progression, and return to sport criteria utilized after operative treatment of a Lisfranc ligament injury.

Level of Evidence: V

Keywords: Closed chain rehabilitation Lisfranc joint, midfoot injury, tarsometatarsal joint

CORRESPONDING AUTHOR

Daniel S. Lorenz

Director of Physical Therapy, Specialists in
Sports and Orthopedic Rehabilitation

7381 W. 133rd St., Suite 302

Overland Park, KS 66213

Phone: (913) 904-1128

Email: danielslorenz@gmail.com

¹ Specialists in Sports and Orthopedic Rehabilitation, Overland Park, KS, USA

² ProSport Physical Therapy, Newport Beach, CA, USA

BACKGROUND AND PURPOSE

Lisfranc joint injuries are the second most common foot injury in athletes, yet are missed or misdiagnosed in 1 of every 5 patients with a foot injury.^{1,2} Midfoot sprains occur in 4% of football players per year, with 29.2% occurring in offensive linemen.¹

The tarsometatarsal joint (TMT) also known as the “Lisfranc complex” consists of the articulation between the midfoot and the forefoot (Figure 1). The 1st through 3rd metatarsals articulate with the corresponding cuneiforms, while the 4th and 5th metatarsals articulate with the cuboid. The transverse arch is formed on the medial side of the foot by the navicular and the 1st metatarsal. Transverse ligaments connect each metatarsal head, however there is no ligamentous coverage on the dorsal surface between the 1st and 2nd metatarsal. The Lisfranc ligament connects the medial cuneiform to the base of the 2nd metatarsal head.²

The diagnosis of a Lisfranc injury is challenging due to the presentation similarities it shares with less severe injuries. Unfortunately, a Lisfranc injury can severely limit athletes in their sport, which is why there is a need for prompt and accurate diagnosis with implementation of evidence based treatment. Midfoot injuries in the athletic population are

either high-energy type consisting of forced hyper-plantarflexion with an additional valgus/varus component, or low energy type consisting of forced hyper-plantarflexion of the midfoot with an axial load through the foot (Figure 2).² The two different forces can cause a wide array of injury severity ranging from mild ligament sprains to displaced fractures. The injuries are difficult to diagnose because of minimal to no diastasis. Even though the low energy types are seemingly less traumatic, they can still limit an athlete from playing at an elite level for several months.

The choice between operative versus non-operative management of this injury starts with the physical exam. Common signs and symptoms seen with Lisfranc injuries are ecchymosis on the plantar surface of the midfoot, midfoot edema, pain with palpation over the midfoot continuing for up to five days after the initial injury,³ and pain with weight-bearing that is typically exacerbated with heel raises.⁴ Special tests include the midfoot squeeze test, forefoot abduction or pronation while maintaining a stable calcaneus, and the “piano key test,” performed when the toes are grasped and moved in a plantar and dorsal direction to elicit pain. If differential diag-

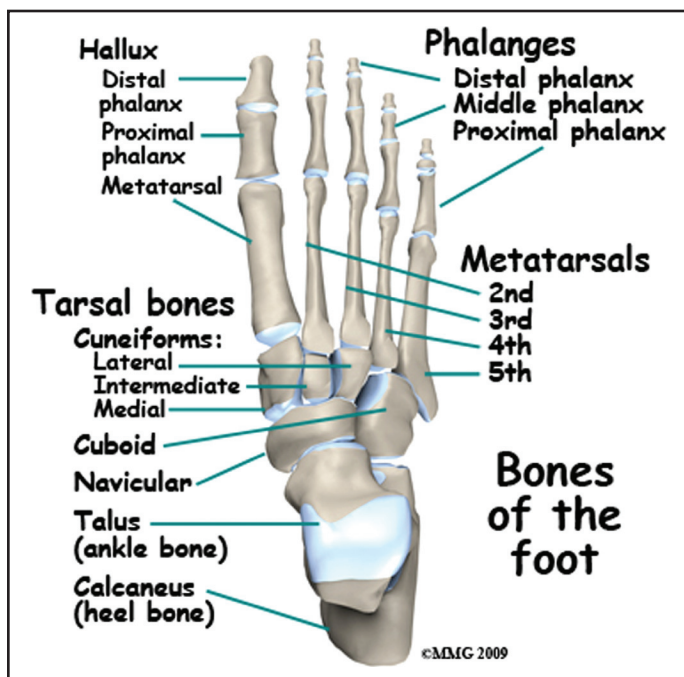


Figure 1. Tarsometatarsal joint anatomy.

Figure courtesy of MMG via <http://www.eorthopod.com>

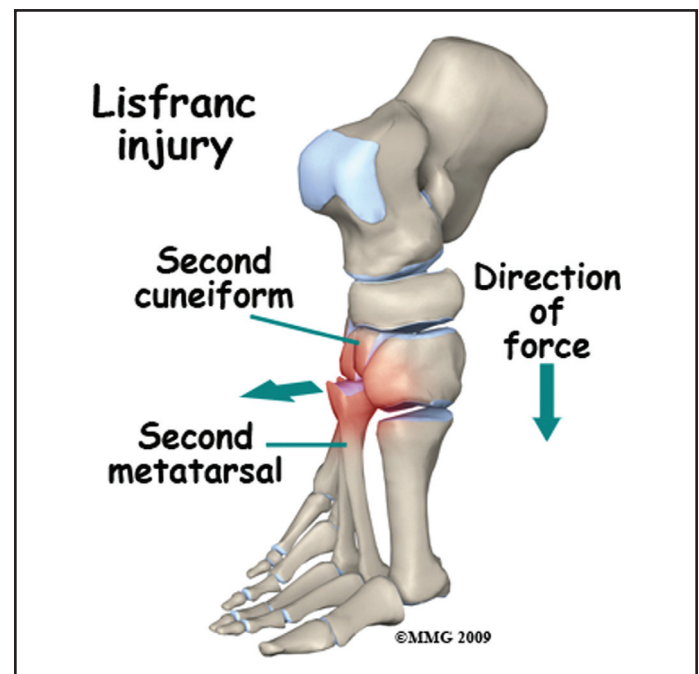


Figure 2. Typical mechanism for a low energy/velocity Lisfranc injury.

Figure courtesy of MMG via <http://www.eorthopod.com>

nosis infers an injury to the Lisfranc complex the patient should be referred for standard radiographic imaging. Standard imaging series includes a weight-bearing AP, and lateral, and 30° oblique foot views.⁵ Nunley and Vertullo⁶ used the following radiographic criteria to assess the Lisfranc ligament complex: 1) distance between the first and second metatarsal bones, 2) distance between the medial and intermediate cuneiform bones on the AP radiograph, 3) continuity of a line formed by the medial border of the second metatarsal bone with the medial border of the intermediate cuneiform bone, 4) the distance between the plantar fifth metatarsal and the plantar medial cuneiform bones on the lateral radiograph, and 5) the presence of avulsion fractures. Images were compared to the contralateral foot, and any diastasis 1 mm greater than that of the uninjured foot was considered diagnostic for this type of injury.⁶ There are three grades used to classify Lisfranc injuries: Grade I sprain to the Lisfranc ligaments with <2 mm diastasis and no arch height loss; Grade II sprains with diastasis of 2-5 mm between the 1st and 2nd metatarsals with no arch height loss; and Grade III sprains with diastasis >5 mm between 1st and 2nd metatarsals with longitudinal arch height loss. For more complex Lisfranc dislocation/fractures, the reader is referred to Myerson et al⁷ for an additional classification system.

Lisfranc injuries can be treated conservatively in some cases. Grade I Lisfranc injuries are considered to be stable and can be treated non-operatively. In a case report, Wadsworth and Eadie⁸ described successful conservative management in a basketball player, highlighting key features including recognition of a prolonged recovery time, adequate rest to allow for soft tissue healing, restoration of a normal gait pattern to prevent chronic overstress of injured tissues, appropriate orthotic prescription, and proprioceptive retraining.⁸ There are a variety of recommendations in the literature for conservative treatment ranging from immediate weight-bearing in an orthotic with arch support to non-weight bearing in a cast for 6 weeks.⁹ Nunley and Vertullo⁶ found that conservative management of Grade I injuries provided excellent results after being casted (non-weight bearing) for six weeks, followed by the use of molded orthotics. Lattermann and coauthors recommend that a repeat radiograph be performed after two weeks of casting



Figure 3. *Leaning heel raises. Leaning places the ankle in more dorsiflexion, making the plantarflexors work over a greater range of motion.*

to ensure that there is no increase in diastasis. If the injury is stable at the two-week re-assessment and the athlete is clinically non-tender, then progression into weight-bearing activities with orthotics may be initiated.⁹ If the injury is not stable, the athlete should remain in the non-weight bearing cast for six weeks.⁶

Contrary to mild Lisfranc injuries that can be managed conservatively, all tarsometatarsal injuries that demonstrate displacement and/or instability require surgical fixation.¹⁰ Clinical outcomes are highly dependent on the restoration of normal anatomic alignment. Current invasive treatment for abnormal anatomic alignment is open reduction and internal fixation (ORIF) using bone screws. Multiple Kirschner wires (K-wires) also have been advocated, but maintaining reduction with them is more difficult.¹¹ In fact, screw fixation has been shown to have significantly greater biomechanical stability than

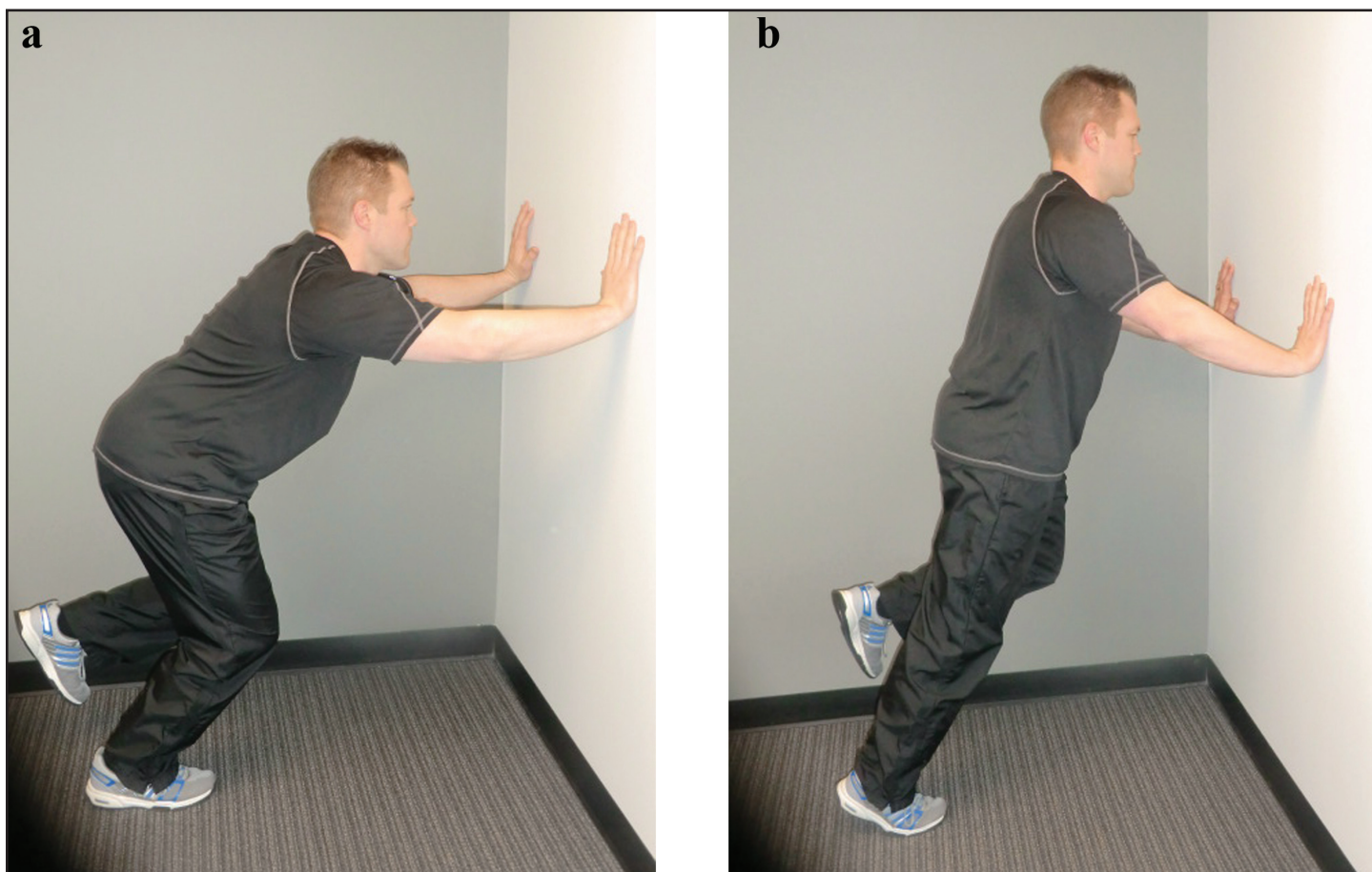


Figure 4. (a) Triple extension heel raise start. (b) Triple extension heel raise end. Movement facilitates combination of ankle plantarflexion, knee extension, and hip extension required for running and jumping.

K-wires.^{12,13} If K-wires are used they should be removed at the six-week post op visit.

Watson and others propose that postoperative patients be placed in a short leg non-weight-bearing cast for the first 3 weeks, then a CAM boot is used for the next 3 to 5 weeks, progressing to full weight bearing by 8 weeks postoperatively.¹⁰ Patients are usually allowed to return to a shoe with an orthosis at 3 months after surgery, and screws are typically removed 4 months postoperatively.¹⁰ If the joint is not fused during surgery, some motion is expected, and this constant motion can cause hardware failure. Watson and colleagues found that if fixation screws remain in place indefinitely, they have a high tendency to break over time, therefore causing pain.¹⁰ At the six-week post op visit, healing is assessed through radiographs. After the 4-6 month check-up, fixation screws are typically removed across the TMT joint and full weight-bearing is allowed as tol-

erated. Thordarson and colleagues¹⁴ reported results from 14 patients with an average follow-up of 20 months. Even though follow-up time ranged from 3-45 months, they concluded that bio-absorbable screws are safe and that they eliminate the need for screw removal. Larger studies with long-term follow-up are needed to determine the true efficacy of such fixation devices.¹⁴

Stable anatomic alignment is the best predictor of outcomes and instability appears to be the primary pain generator.¹⁵ The presence of fractures and/or articular destruction leads to poorer results, regardless of alignment. Incidence of posttraumatic arthritis reportedly ranges from 0-58%.¹⁶ Philbin and colleagues¹⁶ reported that up to 25% of patients developed posttraumatic arthritis even after successful fixation. This same study showed no difference between acute and delayed (>6 weeks) surgical fixation as well as poorer outcomes with purely ligamen-



Figure 5. *Quarter turn on mini-trampoline, two feet.*

tous injuries. Ly and colleagues¹⁷ compared fusion of the first three TMT joints to ORIF and found that the fusion group returned to 92% of their pre-injury activity level compared to the ORIF group who only attained 65% of their pre-injury activity level.

Nunley and Vertullo⁶ described return to play in 15 athletes following either conservative or surgical treatment. Athletes with Grade I injuries had excellent outcomes with conservative treatment and returned to play between 11-18 weeks. Athletes with Grade II injuries had good outcomes with ORIF and returned to play between 12-20 weeks. However, managed conservatively, poorer outcomes were noted with a diastasis >2 mm in Grade II injuries. Results for athletes with Grade III or complex dislocations are not well described, but it is safe to say these injuries would be season ending and have a significantly longer return to play than both Grade I or II injuries.

Based on introductory information contained above, there is a fair amount of information and clear criteria established for conservative vs. operative management of these conditions. However, to the authors knowledge, there are no studies or case reports highlighting the functional progression of an athlete after this injury or the return to play criteria. Therefore, the purpose of this case report is to focus on the exercise, functional progression, and return to sport criteria utilized after operative treatment of a Lisfranc ligament injury.



Figure 6. *Half turn on mini-trampoline, two feet.*

CASE DESCRIPTION: PATIENT HISTORY AND SYSTEMS REVIEW

The subject of this case report was a 14 year-old male (5'9", 160 lb) who injured his left midfoot while wrestling in December of 2011. He had no remarkable history of ankle or foot injury and no medical history pertinent to this case. He describes a mechanism of injury whereby his left knee was flexed and his foot was plantarflexed as he fell backward, with all weight landing on a plantarflexed foot. He felt immediate pain, was unable to walk, and had to be carried off of the wrestling mat. The subject was transported to the emergency room where standard radiographs were performed and his foot was placed in a compressive wrap. Because he was in so much pain, he chose to wear a walking boot he had at home in order to help reduce pain while walking, but eventually the subject had to use bilateral axillary crutches in a non-weight-bearing manner due to pain. He was seen by an orthopedist one week later and non-weight bear-

ing radiographs revealed no signs of injury. A computerized tomography (CT) scan was ordered as the subject continued to have persistent pain. Again, no pathology was discovered. Due to the subject's persistent foot pain, his mother did some research on the internet and requested a weight-bearing radiograph, due to her suspicion of a Lisfranc injury. Almost one month after the injury, the weight-bearing radiograph revealed that the subject did have a Lisfranc injury with a five-millimeter separation. Therefore, the subject underwent an ORIF under fluoroscopic guidance on January 6, 2012 with a 3.5mm screw used to stabilize the joint. He was non-weight bearing for 8 weeks. At 8 weeks post-operatively, he was transitioned to a walking boot "for a limited amount of time", while continuing to use crutches and was progressed to full weight-bearing without crutches approximately at 14 weeks post-operatively. His initial physical therapy evaluation was April 19, 2012. The subjects stated goals were to get back to running and be able to play football in the fall.

Initial Examination

The subject's only complaints at initial evaluation were walking with a limp and pain with running/jogging or when on his toes, as in climbing stairs. At rest pain was 0/10 on a Visual Analog Scale, but increased to 4/10 when he ran or pushed up through his toes. Lower Extremity Functional Scale (LEFS) score was 67/80, with 80/80 being a perfect score with no disability. Dysfunctional areas included the ability to run on even and uneven ground, jumping, hopping, and making sharp turns. The LEFS is a reliable and valid indicator of lower extremity function.¹⁸

Gait was observed to be antalgic with lack of push off at terminal stance and he demonstrated a compensated Trendelenburg pattern. He was unable to walk on his toes because of weakness and pain. Ankle range of motion was symmetrical bilaterally in open chain, but he lacked 10° of dorsiflexion in closed chain when measured with a goniometer. There was minimal swelling in the midfoot compared to the right ankle, and his figure-of-eight measurement was within 1 cm. In sitting, ankle plantarflexion strength via hand held dynamometer was 65% of the right, while dorsiflexion and inversion strength was 83% and 85%, respectively, as compared to the right. There was no deficit in eversion. Left gluteus medius

strength was 4/5 tested in side lying with synergistic dominance noted of the psoas major. STAR Excursion Balance Test (SEBT) revealed a deficit of 15 centimeters in both anterior reach and posterior-medial reach when in left lower extremity stance as compared to the right. The SEBT has previously been found to be a valid and reliable measure of function, dynamic balance, and predictor of injury risk.¹⁹⁻²¹ The subject had been performing open chain knee extensions at his school gym in order to maintain his quadriceps strength, therefore, no atrophy was noted and strength was normal via manual muscle test.

Interventions after first visit

The subject was placed back on a single crutch in order to help normalize gait on the first visit. Home exercise program included elastic band exercises for ankle strengthening, standing bilateral heel raises, and exercises to strengthen the gluteus maximus and medius. Additionally, he was instructed in stretches for the gastrocnemius-soleus complex, the hamstrings, quadriceps, and hip flexors.

Interventions and the Rehabilitation Plan

Over the eight weeks after the initial visit, the subject had physical therapy two times per week following the progression outlined in Table 1. In addition to the strengthening exercises listed in Table 1 for closed chain plantarflexion strength, he also completed various lower extremity strengthening exercises such as lunges and step-ups, as well as single leg balance activities on stable and unstable surfaces. To restore his closed chain dorsiflexion, posterior talar glide mobilizations with movement as described by Vincenzino et al²² were performed each session (three sets of twenty mobilizations) until SEBT was symmetrical and dorsiflexion measure was symmetrical via goniometric measurements in closed chain, both of which were achieved by the third session. All sessions ended with ice massage over the foot until subjective numbness was achieved. Additionally, he was instructed to utilize ice massage at home at least three times per day and after each home exercise session.

Approximately three weeks after the initial evaluation, the subject was able to perform single leg heel raises pain free. His strength was re-tested with a hand-held dynamometer and he demonstrated symmetrical force production in inversion, eversion,

Table 1. *Progression of plantarflexion strengthening and restoration of power. The athlete did not progress to the next step until each exercise was completed pain free and with proper technique (i.e. no hip hiking to complete heel raises). All strengthening exercises were performed 3-4 sets of 15-25 repetitions.*

Bilateral heel raises in standing
Bilateral heel raise, single leg eccentric lower
Single leg heel raise from standing
Bilateral leaning heel raises (Figure 3)
Bilateral leaning heel raises, single leg eccentric lower
Single leg leaning heel raises
Single leg triple extension heel raises (Figure 4a, 4b)
Mini-tramp low impact exercises (1:2 work:rest ratio)
- Bilateral 2-legged jumps in place - ¼ turns in place (2 legs) (Figure 5) - ½ turns in place (2 legs) (Figure 6) - Jog in place - 3 hops uninvolved, 1 hop involved - 2 hops uninvolved, 2 hops involved - 1 hop uninvolved, 3 hops involved
Agility ladder
- Various 2 foot sagittal, frontal and transverse plane patterns - “Hopscotch” to involved side (2 feet to 1 foot)
Single leg A/P hops in place (all in place hops 1:2 work:rest ratio)
Single leg M/L hops in place
Single leg transverse plane hops in place
Single leg hops in agility ladder, multi-plane

plantarflexion, and dorsiflexion. Low impact exercises on a mini-tramp were initiated 3 weeks after therapy commenced, on his 6th visit. They were performed in a 1:2 work:rest ratio, starting with 15 seconds of work time and building to 30 seconds. Once mini-tramp exercises were performed without pain, he was transitioned to an agility ladder where he progressed from double leg hops (8th visit) to “hopscotch” patterns (two feet to one foot) and then to one-legged hops in place (10th visit). Once one-leg hops in place

were pain free, he was permitted to start jogging and to perform single leg patterns in the agility ladder (12th visit). He started with intervals on a treadmill with a 1:2 work:rest ratio at a comfortable pace. With each session of therapy, work:rest ratio increased. Sessions closed with walking on a treadmill at a normal walking pace for ten minutes with an incline of 10% to facilitate push-off. Once jogging forward and backward was pain free and without a limp, 45° cuts, shuffling, carioca drills, and 90° cuts were pro-

gressively added as long as he had no pain with any progression. Eight weeks after starting therapy on his 16th visit, he was hop tested. Single leg hop, triple hop, and 6-meter timed hop were 92%, 74%, and 86% of the uninjured, respectively. Based on these scores, the subject was advised to emphasize single leg plyometric activities and rope jumping as part of his home program to help improve single leg power. He was to start a summer strength and conditioning program with his football team the following week and he was allowed to do so with pain as the only restriction from participation.

The subject started conditioning with his football team during summer workouts in early July, approximately 10 weeks after the start of therapy, and did not experience pain with conditioning activities that included sprints of various distances. At this time, he transitioned to one day of physical therapy per week and he continued with a supervised home program. Approximately one week after the start of football conditioning, he came back to therapy reporting that he felt like his screw might be loosening because it looked like something was “poking out” to him. He was still not experiencing any pain and was able to participate in his conditioning, but the protrusion was tender to touch and was irritated by his shoe. The subject and his mother were advised to follow-up with their orthopedist and refrain from all activity. One week later, the screw was removed. The subject was allowed to weight-bear immediately after the screw was removed, and was allowed to lift weights, but was instructed not to do any running or jumping for two weeks.

Two and a half weeks post-screw removal, he returned to therapy for re-evaluation. His gait was re-evaluated as well as his ability to hop in multiple planes on a single leg. He had no complaints of pain or visible deficits. Therefore, hop testing was performed with the following results: single leg hop, triple hop, and 6-meter timed hop were 98%, 100%, and 91% respectively, of the uninjured. The LEFS was repeated and the subject scored an 80/80. At that time, he was allowed to return to full participation in team weight training and conditioning workouts and a follow-up appointment was scheduled for three weeks prior to the start of fall practice if there were any issues. Due to insurance constraints, specific follow-up regarding his progression during this

timeframe was a challenge. He was instructed to run every other day, provided he had no pain prior to the workout, for the first two weeks. After the two-week period, he was allowed to run every day. If he had pain the following day after a workout in his foot, he was advised not to run again until he was pain free. The subject had no issues however and required no modifications to this plan. He was able to start fall practice in August and completed his season without any issues or re-injury of the left foot.

DISCUSSION

There are several key elements of this case that are worthwhile to the sports physical therapist. First of all, an accurate diagnosis is essential. The sports physical therapist should be aware that midfoot pain should warrant weight-bearing radiographs to rule out a Lisfranc injury, particularly if the athlete reports the mechanism and is not responding to conservative care.

Next, this case highlights how important it is to test an athlete in functional patterns beyond just the ability to walk or run. The physician in this case felt that physical therapy was not needed because the subject outwardly had a normal gait from his perspective. Although he had a slightly antalgic gait on initial exam, the profound deficits in closed-chain plantarflexion were not manifested until a complete, focused functionally-based evaluation was performed. Furthermore, as he progressed through his rehabilitation program, he was running with a normal gait and participating in 7-on-7 drills with his teammates, but this athlete continued to display a significant deficit of single leg power.

The main purpose of this case presentation is to highlight an exercise progression to restore closed chain plantarflexion strength and single limb functional abilities in an athlete with a foot pathology, in order to promote return of running and jumping function. There is a lack of evidence regarding functional foot outcome measures that have been proven to be valid and reliable for determining an athlete's return to play readiness following a Lisfranc injury. Therefore, the use of functional testing was paramount, and is a necessary part of determining the readiness of an athlete to return to their sport. The majority of investigations regarding assessment of the athlete's

readiness for safe return to sport involves ACL injury/deficiency and reconstruction,²³⁻³¹. After appropriate strength has been successfully restored, various hop tests for the lower extremities have been advocated and proven to be a reliable indicator of functional strength.^{23-29,31} Hopping is a common theme in the large majority of these tests as it measures primarily concentric and eccentric power, eccentric control, balance, and lower extremity stabilization ability. Power is a common parameter in many sports, so it was determined that hop testing would be appropriate in this case. The crossover hop test was deferred in this case because the physical therapist felt that medial/lateral stability was not an issue with this subject; however, the single leg hop, triple hop, and 6-meter timed hop were used to assess single leg strength and power. After the first assessment using the hop tests, it was evident that the subject lacked single leg power. Therefore, the focus of his workouts shifted to single leg power activities for the month following the results of the first hop tests.

Additionally, this case highlights that screw removal was necessary. Without having this done, it would have limited his functional ability. It is plausible that the subject's activity level caused the screw to loosen. Once the screw was removed, he did exceedingly well and progressed quickly. Although data is limited regarding whether or not the screw should be removed, it is evident that in this case that it was necessary. There appeared to be no adverse effects from removing the screw as his hop tests maintained their symmetry during re-testing. Therefore, it appears screw removal should be on a case-by-case basis and based on pain, dysfunction, or physical appearance of screw protrusion. Given that this is a single subject case report, no definitive conclusions can be drawn about this issue.

Finally, this case provided a framework for the sports physical therapist to utilize when designing rehabilitation programs for an athlete who is recovering from a Lisfranc injury. Exercises to promote return of plantarflexion strength were systematic and progressive, and his return to impact activities followed a safe, logical progression. All exercises were performed on two feet, progressed to eccentric only on the involved, to single leg, then a gradual progression of impact activities. Since the triple and 6-meter timed hop were at a deficit compared to the

involved limb, plyometric activities were the emphasis of his sessions in the terminal phases of rehabilitation. Restoration of his single leg power was effectively his return to play criteria and the summer conditioning workouts afforded him the time to progressively overload the foot prior to the start of the football season. The authors note one modality that was not used until the end of his treatment was aquatic therapy. Because of the time of year of the subject's rehabilitation in a Midwestern city with limited resources available, there was no opportunity to use a pool. Ideally, given the nature of this injury, he would have been an ideal candidate to supplement land-based therapy with aquatic therapy. Aquatic therapy has been previously advocated as a useful tool for lower extremity rehabilitation in the athlete.³²⁻³⁴ Once the weather was more appropriate eight weeks after the start of therapy, he supplemented his land-based workouts with independent pool exercises on off-days that included running, shuffling, cariocas, high knees, "power skips," counter movement jumps, and single leg hops and jumps. Future cases could highlight a rehabilitation program that combines land-based and aquatic-based progression for foot injuries. Finally, specific functional testing criteria for athletes recovering from foot injuries should be established that may include a return to running/sprinting progression as well.

REFERENCES

1. Meyer SA, Callaghan JJ, Albright JP, et al: Midfoot sprains in collegiate football players. *Am J Sports Med.* 1994; 22: 392-401.
2. Burroughs KE, Reimer CD, Fields KB. Lisfranc injury of the foot: a commonly missed diagnosis. *Am Fam Physician.* 1998 Jul;58(1):118-24.
3. Englanoff G, Anglin D, Hutson HR. Lisfranc fracture-dislocation: a frequently missed diagnosis in the emergency department. *Ann Emerg Med.* 1995;26: 229-33.
4. Mantas JP, Burks RT. Lisfranc injuries in the athlete. *Clin Sports Med.* 1994;13:719-30.
5. Shapiro MS, Wascher DC, Finerman GA. Rupture of Lisfranc's ligament in athletes. *Am J Sports Med.* 1994; 22: 687-691.
6. Nunley J, Vertullo C. Classification, Investigation, and Management of Midfoot Sprains: Lisfranc injuries in the athlete. *Am J Sports Med.* 2002;30(6):871-878.

7. Myerson MS, Fisher RT, Burgess AR, et al. Fracture dislocations of the tarsometatarsal joints: end results correlated with pathology and treatment. *Foot Ankle*. 1986; 6:225-242.
8. Wadsworth DJ, Eadie NT. Conservative management of a subtle Lisfranc injury: a case report. *J Orthop Sports Phys Ther*. 2005; 35(3): 154-164.
9. Lattermann C, Goldstein JL, Wukich DK, Lee S, Bach BR. Practical Management of Lisfranc Injuries in Athletes. *Clin J Sport Med*. 2007;17:311-315.
10. Watson TS, Shurnas PS, Denker J. Treatment of Lisfranc joint injury: current concepts. *J Am Acad Orthop Surg*. Dec 2010;18(12):718-28.
11. Smith SE, Camasta CA, Cass AD. A technique for isolated arthrodesis of the second metatarsocuneiform joint. *J Foot Ankle Surg*. Sep-Oct 2009;48(5):606-11.
12. Gaines RJ, Wright G, Stewart J. Injury to the tarsometatarsal joint complex during fixation of Lisfranc fracture dislocations: an anatomic study. *J Trauma*. Apr 2009;66(4):1125-8.
13. Lee CA, Birkedal JP, Dickerson EA, et al. Stabilization of Lisfranc joint injuries: a biomechanical study. *Foot Ankle Int*. May 2004;25(5):365-70.
14. Thordarson DB, Hurvitz G. PLA screw fixation of Lisfranc injuries. *Foot Ankle Int*. Nov 2002;23(11):1003-7.
15. Nithyananth M, Boopalan PR, Titus VT, Sundararaj GD, Lee VN. Long-term outcome of high-energy open Lisfranc injuries: a retrospective study. *J Trauma*. Mar 2011;70(3):710-6.
16. Philbin T, Rosenberg G, Sferra JJ. Complications of missed or untreated Lisfranc injuries. *Foot Ankle Clin*. Mar 2003;8(1):61-71.
17. Ly TV, Coetzee JC. Treatment of primarily ligamentous Lisfranc joint injuries: primary arthrodesis compared with open reduction and internal fixation. A prospective, randomized study. *J Bone Joint Surg Am*. Mar 2006;88(3):514-20.
18. Binkley JM, Stratford PW, Lott SA, Riddle DL. The Lower Extremity Functional Scale (LEFS): scale development, measurement properties, and clinical application. North American Orthopaedic Rehabilitation Research Network. *Phys Ther*. 1999; 79: 371-83.
19. Kivlan BR, Martin RL. Functional performance testing of the hip in athletes: a systematic review of reliability and validity. *Int J Sports Phys Ther*. 2012; 7: 402-12.
20. Plisky PJ, Rauh MJ, Kaminski TW, Underwood FB. Star Excursion Balance Test as a predictor of injury in high school basketball players. *J Orthop Sports Phys Ther*. 2006; 36: 911-19.
21. Kinzey SJ, Armstrong CW. The reliability of the star-excursion test in assessing dynamic balance. *J Orthop Sports Phys Ther*. 1998; 27: 356-60.
22. Vicenzino B, Branjerdporn M, Teys P, Jordan K. Initial changes in posterior talar glide and dorsiflexion of the ankle after mobilization with movement in individuals with recurrent ankle sprain. *J Orthop Sports Phys Ther*. 2006;36(7):464-71.
23. Alkjaer T, Simonsen EB, Magnusson SP, et al. Differences in the movement pattern of a forward lunge in two types of anterior cruciate ligament deficient patients: copers and non-copers. *Clin Biomech*. 2002; 17: 586-593.
24. Augustsson J, Thomee R, Karlsson J. Ability of a new hop test to determine functional deficits after an ACL reconstruction. *Knee Surg Sports Traumatol Arthrosc*. 2004; 12(5): 350-356.
25. Colby SM, Hintermeister RA, Torry MR, Steadman RJ. Lower limb stability with ACL impairment. *J Orthop Sports Phys Ther*. 1999;29:444-454.
26. Itoh H, Durosaka M, Yoshiya S, et al. Evaluation of functional deficits determined by four different hop tests in patients with anterior cruciate deficiency. *Knee Surg Sports Traumatol Arthrosc*. 1998; 6: 241-245.
27. Petschnig R, Baron R, Albrecht M. The relationship between isokinetic quadriceps strength test and hop tests for distance and one-legged vertical jump test following anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther*. 1998;28:23-31.
28. Pfeifer K, Banzer W. Motor performance in different dynamic tests in knee rehabilitation. *Scan J Med Sci Sports*. 1999;9:19-27.
29. Reid A, Birmingham TB, Stratford PW, Alcock GK, Giffin JR. Hop testing provides a reliable and valid outcome measure during rehabilitation after anterior cruciate ligament reconstruction. *Phys Ther*. 2007;87:337-349.
30. Risberg MA, Ekeland A. Assessment of functional tests after anterior cruciate ligament surgery. *J Orthop Sports Phys Ther*. 1994; 19(4): 212-217.
31. Wilk KE, Romaniello WT, Soscia SM, Arrigo CA, Andrews JR. The relationship between subjective knee scores, isokinetic testing, and functional testing in the ACL reconstructed knee. *J Orthop Sports Phys Ther*. 1994;20:60-73.
32. Thein JM, Brody LT. Aquatic-based rehabilitation and training of the elite athlete. *J Orthop Sports Phys Ther*. 1998; 27: 32-41.
33. Prins J, Cutner D. Aquatic therapy in the rehabilitation of athletic injuries. *Clin Sports Med*. 1999; 18: 447-61.
34. Becker BE. Aquatic therapy: scientific foundations and clinical rehabilitation applications. *Phys Med Rehabil*. 2009; 1: 859-72.